Corporate Governance Reform and the Korea Discount: Lessons from Japan's Reform Experience

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Abstract

South Korean equities have traded at a persistent structural discount to developed-market peers for over two decades. Using a 20-year monthly panel of index-level price-to-book (P/B) ratios for KOSPI, TOPIX, S&P 500, and MSCI EM from 2004 to 2024, we document that the Korea Discount averages $-0.177\times$ relative to TOPIX ($t = -3.23$; 95% CI: $[-0.284\times, -0.069\times]$) and $-0.601\times$ relative to MSCI EM ($t = -10.30$). We use Japan's three staged corporate governance reforms — the Stewardship Code (February 2014), the Corporate Governance Code (June 2015), and the Tokyo Stock Exchange P/B reform (March 2023) — as policy benchmarks rather than as clean natural experiments. A descriptive event-window analysis of the KOSPI–TOPIX P/B spread shows large movements around the reform dates, but the implemented saturated cohort-by-month design does not support conventional standard errors or causal inference. Panel OLS with country and time fixed effects produces imprecise reform-by-Japan interactions: wild-bootstrap p-values are 0.625, 0.375, and 0.500 for the three reforms, respectively, given only $N = 4$ country clusters. A demeaned synthetic-control exercise for the 2023 TSE P/B reform assigns full weight to MSCI EM Asia (RMSPE = 0.1451); the post-reform gap does not provide robust evidence of an idiosyncratic Japan governance premium after accounting for regional equity momentum. The evidence therefore supports the existence of a persistent Korea Discount, but it does not by itself identify the valuation gain Korea would

realize from adopting Japan-style reforms.

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1 Introduction

South Korean equities occupy an anomalous position in global capital markets. Despite Korea's membership in the G20, its well-developed export-oriented industrial base, and macroeconomic fundamentals broadly comparable to other high-income economies, KOSPI Composite index-level price-to-book ratios have persistently trailed those of comparable benchmarks for over two decades. Over the 2004–2024 sample period, the KOSPI P/B ratio averaged $-0.177\times$ below TOPIX ($t = -3.23$, 95% CI: $[-0.284\times, -0.069\times]$) and $-0.601\times$ below MSCI EM ($t = -10.30$, 95% CI: $[-0.716\times, -0.486\times]$). This persistent, statistically significant undervaluation — the "Korea Discount" — remains one of the most debated puzzles in international equity valuation. Figure 1 displays the full 20-year trajectory of KOSPI P/B relative to its benchmark peers.

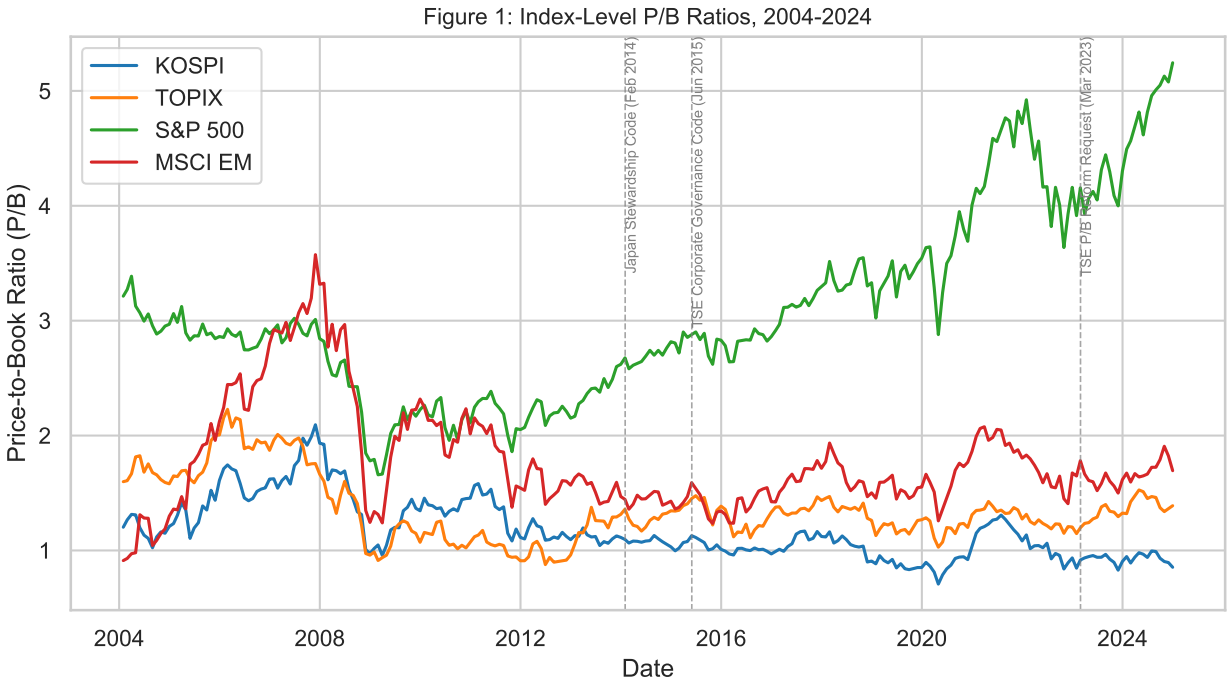

Figure 1: Index-Level Price-to-Book Ratios, 2004–2024. KOSPI trades at a persistent discount to TOPIX, S&P 500, and MSCI EM throughout the sample period. The Korea Discount averages $-0.177\times$ relative to TOPIX ($t = -3.23$) and $-0.601\times$ relative to MSCI EM ($t = -10.30$). Dashed vertical lines mark Japan's three governance reform dates (February 2014, June 2015, March 2023).

We emphasize three structural channels that plausibly account for the discount: (1) chaebol cross-shareholding networks that generate governance opacity and suppress minority-shareholder

value; (2) a regulatory environment that historically afforded limited legal recourse to outside investors; and (3) a geopolitical risk premium arising from North Korea's nuclear program. Japan provides a useful policy benchmark, but not a clean source of causal identification in the current data. Japan faced an analogous valuation discount through much of the 2000s and 2010s and responded with three staged governance reforms — the Stewardship Code (February 2014), the Corporate Governance Code (June 2015), and the Tokyo Stock Exchange P/B reform (March 2023). Korea did not implement an equally forceful staggered program over the primary 2004–2024 sample, but Korea's own Stewardship Code and 2024 Value-Up Program mean that a strict never-treated interpretation would overstate the design.

This paper makes three contributions. First, we construct the first panel study using all three Japan governance reform dates jointly as descriptive event windows for the KOSPI–TOPIX valuation spread. This exercise documents timing and magnitude, while making explicit that the saturated design and overlapping 2014–2015 windows do not deliver valid conventional inference. Second, we report two-way fixed-effects panel estimates with small-cluster wild-bootstrap p-values, showing that the reform interactions are too noisy to sustain strong directional claims. Third, we provide a synthetic-control check for the 2023 TSE P/B reform and transparently report the single-donor weight structure, which limits the strength of the counterfactual interpretation.

Section 2 surveys the prior literature and institutional background. Section 3 describes data, causal mechanisms, and identification strategy. Section 4 reports empirical results. Section 5 discusses limitations, policy implications, and conclusions. Appendices contain variable definitions and robustness tables.

2 Literature Review and Institutional Background

2.1 The Korea Discount: Prior Evidence and the Chaebol Structure

The Korea Discount has attracted a substantial body of academic and practitioner research. South Korean corporate structure is dominated by *chaebols* — large, family-controlled conglomerates that

collectively account for a substantial share of KOSPI market capitalization and national exports. The defining feature of the chaebol is a layered ownership architecture in which a founding family holds a relatively small direct equity stake in each affiliate but exercises effective control through a web of cross-shareholdings and pyramid ownership (Claessens et al., 2000). This generates a *principal–principal* agency problem that is analytically distinct from the classic principal–agent framework of Jensen and Meckling (1976): the conflict is between the controlling family (a concentrated principal) and minority shareholders (diffuse outside principals), with the family having both the incentive and means to extract private benefits — tunneling — without bearing the proportionate economic cost.

Baek et al. (2004) provide the foundational empirical paper on this mechanism. Exploiting the 1997 Korean financial crisis as an exogenous shock, they show that firms with greater separation between control rights and cash-flow rights experienced larger equity value declines, establishing that tunneling has quantifiable and economically large effects on market valuations. Claessens et al. (2000) document systematically across East Asian corporations that the median Korean firm has a control-to-cash-flow ratio of approximately 2:1, highest in the region, and that this separation is associated with a significant discount to Tobin's q.

Black et al. (2006) construct a Korean corporate governance index (KCGI) using survey data on board practices, disclosure, shareholder rights, and ownership structure, and show via panel OLS that higher KCGI scores are associated with significantly higher Tobin's q values — results that survive controls for firm size, leverage, profitability, and industry fixed effects. Korea Capital Market Institute (2023) provide the most recent comprehensive assessment from the Korea Capital Market Institute, identifying governance structure, low shareholder returns, market accessibility deficiencies, and the inheritance tax structure (rates up to 60%, applied to market value, creating controlling-family incentives to keep share prices depressed) as the four primary contributors.

An important methodological distinction separates prior work from this paper. The firm-level literature identifies the cross-sectional governance–valuation relationship using firm-level governance indices, but cannot readily estimate the causal effect of a nationwide governance reform

shock. Our event-study and panel designs fill this gap, at the cost of operating with a single treated country ($N = 1$) and the inferential challenges this entails.

2.2 Japan's Governance Reforms and Their Empirical Literature

Japan's governance reform program is the source of identifying variation for this study. Japan faced a structurally similar valuation discount through the 2000s and 2010s, attributed to cross-shareholding (*keiretsu* networks), weak board independence, and low dividend payout ratios. The government responded with three landmark interventions.

Japan Stewardship Code (February 2014). The Financial Services Agency introduced Japan's Stewardship Code in February 2014, requiring institutional investors to adopt a governance engagement policy and disclose voting rights exercise. The comply-or-explain framework achieved high adoption among domestic and eventually foreign managers with Japanese equity mandates. Miyajima and Saito (2023) document that the Code accelerated the unwinding of cross-shareholding relationships: institutional investors now under formal obligation to justify passive holdings of cross-held shares began pressing companies to restructure these positions. Firms with larger cross-holding reductions experienced commensurately larger improvements in Tobin's q.

Corporate Governance Code (June 2015). The Tokyo Stock Exchange and FSA jointly introduced the Corporate Governance Code in June 2015, requiring all TSE-listed companies to comply with governance principles or explain deviations. The code mandated at least two independent outside directors — a fundamental change for Japanese boards historically composed almost entirely of long-serving inside managers (Eberhart, 2012). By 2017, over 90% of JPX-Nikkei 400 constituents had appointed at least two independent directors. Eberhart (2012) finds that transitions toward independent-director-inclusive boards are associated with improvements in operating performance and, over longer horizons, equity valuations.

TSE P/B Reform (March 2023). The Tokyo Stock Exchange's March 2023 "Request for Disclosure of Policies on Capital Cost and Stock Price Awareness" instructed companies trading below 1.0× book value to disclose specific plans to improve capital efficiency. Its impact was

immediate: TOPIX P/B rose from approximately 1.0× in early 2023 to 1.36× over the 2023–2024 post-reform period, while KOSPI P/B remained at approximately 0.93× — widening the KOSPI–TOPIX spread.

All three reform dates are locked in `config.py` before any data transformation, providing a look-ahead bias firewall: Stewardship Code 2014-02-01, Corporate Governance Code 2015-06-01, TSE P/B reform 2023-03-01.

Korea's own policy response is instructive for comparison. Korea adopted a Stewardship Code in December 2016 (revised 2018), but it was widely assessed as weaker in enforcement than Japan's 2014 equivalent — the National Pension Service's engagement remained intermittent and the comply-or-explain mechanism produced limited behavioral change. The *Korea Corporate Value-Up Program* (launched by the FSC in 2024) is modeled explicitly on the 2023 TSE P/B reform but remains voluntary; the FSC has proposed Commercial Act amendments strengthening fiduciary duties to explicitly include minority shareholders. This reform trajectory makes Korea a partial comparison case rather than a strictly never-treated unit.

2.3 Corporate Governance, Legal Institutions, and Equity Valuation

The broader governance–valuation literature provides the theoretical grounding for the Korea Discount hypothesis. Shleifer and Vishny (1997) survey the corporate governance literature and document that ownership concentration, while potentially reducing manager–shareholder conflicts, intensifies conflicts between controlling shareholders and minority investors — exactly the principal–principal problem that characterizes chaebols. Legal protection of minority shareholders is established as a first-order determinant of equity valuations across countries.

La Porta et al. (1998) and La Porta et al. (2002) establish the foundational law-and-finance framework: countries with stronger statutory investor protections have larger, more liquid capital markets. La Porta et al. (2002) show that investor protection is causally associated with higher dividend yields and higher Tobin's q. The Korea Discount is a specific application of this framework: Korea's civil-law tradition combined with the chaebol structure places it toward the weak-investor-

protection end of the international distribution. Japan's reform program demonstrates that de facto governance practices can be improved through targeted regulatory intervention even within a civil-law system. Gompers et al. (2003) provide corroborating evidence in the US context, showing that firms with stronger shareholder rights earn significantly higher abnormal returns.

Specific legal deficiencies during the primary sample period in Korea include: high ownership thresholds for minority shareholders to bring derivative suits; comparatively lax board independence requirements relative to OECD peers (the Commercial Act did not mandate a majority of independent directors until reforms beginning in 2020); related-party transaction disclosure thresholds that permitted intragroup transactions without shareholder approval; and shareholder proposal mechanisms requiring higher ownership thresholds than the US or UK.

2.4 Geopolitical Risk

A second structural channel is the country-specific risk premium arising from North Korea's nuclear weapons and ballistic missile program. North Korea conducted six nuclear tests over the sample period: October 2006, May 2009, February 2013, January 2016, September 2016, and September 2017. The 2017 test, alongside ICBM launches demonstrating continental US range, coincided with the largest single-period widening of the Korea–TOPIX P/B spread visible in Figure 1. Unlike the governance channels, which in principle could be eliminated through domestic policy reform, the geopolitical risk channel reflects an external threat with no direct Korean government lever for short-term resolution.

The Jeong et al. (2021) GPRNK analysis finds that North Korean provocations primarily affect systematic risk and market beta — a level effect on the required return — rather than causing permanent cumulative discount widening following each escalation event. This suggests the geopolitical risk premium is embedded in the structural discount level rather than generating time-series variation. The empirical results in Section 4 confirm this interpretation: the GPR escalation coefficient is statistically insignificant at conventional levels.

2.5 Empirical Methods and Inference

The empirical design is intentionally conservative. Staggered difference-in-differences methods such as Cengiz et al. (2019) and Baker et al. (2022) motivate the concern that pooled two-way fixed-effects estimators can be misleading when treatment effects differ across cohorts. In this application, however, the data contain only one reforming country and three events within that same country, with overlap between the 2014 and 2015 reform windows. The event-window estimates are therefore presented as descriptive abnormal movements in the KOSPI–TOPIX spread, not as a clean stacked DiD estimator.

The synthetic control of Abadie et al. (2010) is designed for settings with a single treated unit, but its credibility depends on pre-treatment fit, donor-pool plausibility, and a stable weight structure. Wild-bootstrap inference (Cameron et al., 2008) with Rademacher weights is used for the panel OLS interactions because asymptotic cluster-robust variance estimators perform poorly with only $N = 4$ country clusters. Even the wild bootstrap is limited in such a small panel, so we report p-values alongside point estimates and avoid interpreting statistically insignificant estimates as evidence of zero treatment effects.

3 Methodology

3.1 Data Sources and Variable Construction

The primary data are monthly index-level price-to-book (P/B) ratios exported from the Bloomberg Terminal using the Bloomberg Data License Historical Data Request protocol (field code: `PX_TO_BOOK_RATIO`), which reports aggregate market capitalization divided by aggregate book value of equity across all index constituents, rebalanced at the index's standard reconstitution frequency.

Four benchmark indices constitute the core panel: KOSPI Composite (Bloomberg ticker: KOSPI Index), TOPIX (TPX Index), S&P 500 (SPX Index), and MSCI EM (MXEF Index). The primary analysis sample spans January 2004 to December 2024 — 252 calendar months per series —

yielding 1,008 country-month observations. The processed panel stored on disk includes additional observations through April 2026, for 1,072 rows, because the raw extracts were updated after the primary paper window. All Bloomberg raw CSV exports are version-controlled in `data/raw/` with provenance documented in `data/raw/MANIFEST.md` (vintage date: 2026-04-20).

We prefer P/B over price-to-earnings (P/E) for three reasons: (i) book value is considerably less cyclically volatile than earnings, making P/B a more stable measure of structural discount; (ii) book value does not go negative for index-level aggregates, avoiding the sign-change problem during recessions; and (iii) the TSE P/B reform and Korea's Value-Up Program are explicitly framed in P/B terms. P/E robustness results are presented in Appendix B; results are qualitatively consistent but noisier.

The canonical panel has schema (`date`, `country`, `pb`, `pe`, `fx_rate`), stored as `data/processed/panel.parqu` The Geopolitical Risk index (Caldara and Iacoviello, 2022), with the Korea-specific GPRNK sub-index (Jeong et al., 2021) isolating North Korean escalation events, is sourced at monthly frequency and complete over the full sample period. All four monthly P/B series are complete with no gaps.

The use of official, published index-level P/B series substantially mitigates survivorship bias concerns: P/B ratios are computed by the index provider over all constituents as of each month-end, including firms that subsequently exit. Index constituent turnover is a potential source of composition bias if lower-valuation firms are systematically replaced by higher-valuation entrants, but the Korea Discount has *persisted* despite constituent turnover, and the 2008–2009 global financial crisis P/B compression affects all four series simultaneously in a manner consistent with known economic events — providing a cross-market consistency check.

3.2 Causal Mechanisms

The Korea Discount reflects three compounding structural channels, each of which independently suppresses the equity valuation that international investors assign to KOSPI.

Chaebol Opacity and Agency Costs. Chaebol-affiliated firms represent the substantial majority of KOSPI market capitalization. The principal–principal agency problem arising from the

Table 1: Summary statistics of index-level P/B ratios by country and sub-period.

country	Period	mean	median	std	min	max
KOSPI	Full (2004–2024)	1.18	1.10	0.26	0.71	2.09
MSCI_EM	Full (2004–2024)	1.78	1.65	0.47	0.91	3.58
SP500	Full (2004–2024)	3.08	2.89	0.81	1.66	5.25
TOPIX	Full (2004–2024)	1.35	1.31	0.28	0.88	2.23
KOSPI	Pre-reform (2004–2013)	1.36	1.34	0.25	0.96	2.09
MSCI_EM	Pre-reform (2004–2013)	1.97	1.91	0.60	0.91	3.58
SP500	Pre-reform (2004–2013)	2.51	2.42	0.40	1.66	3.39
TOPIX	Pre-reform (2004–2013)	1.41	1.29	0.39	0.88	2.23
KOSPI	Reform era (2014–2022)	1.02	1.03	0.12	0.71	1.31
MSCI_EM	Reform era (2014–2022)	1.59	1.55	0.20	1.24	2.08
SP500	Reform era (2014–2022)	3.39	3.28	0.63	2.58	4.92
TOPIX	Reform era (2014–2022)	1.29	1.28	0.10	1.03	1.48
KOSPI	Post-TSE (2023–2024)	0.93	0.94	0.04	0.83	1.00
MSCI_EM	Post-TSE (2023–2024)	1.66	1.66	0.10	1.50	1.91
SP500	Post-TSE (2023–2024)	4.51	4.47	0.43	3.91	5.25
TOPIX	Post-TSE (2023–2024)	1.36	1.35	0.10	1.15	1.53

control-to-cash-flow wedge generates three distinct opacity mechanisms: intragroup related-party transactions that are difficult to assess from public disclosures; group-level accounting that obscures individual entity economics; and controlling-family incentives to keep equity prices depressed during generational wealth transfers, given inheritance tax rates up to 60% applied to market value. The "double-counting" problem in index-level valuation — where both parent companies and multiple listed subsidiaries appear in KOSPI — adds a mechanical book-value inflation that international investors must discount.

Minority Shareholder Recourse Deficit. The legal deficiencies described in Section 2 collectively reduce the expected return that minority shareholders can extract from KOSPI-listed firms. Following La Porta et al. (1998, 2002), weak statutory investor protections are associated with lower equity valuations across countries. Korea's position toward the weak-protection end of the international distribution during 2004–2024 generates a commensurate valuation penalty.

Geopolitical Risk Premium. The tail risk of armed conflict on the Korean peninsula warrants a discount to intrinsic value not shared by TOPIX, S&P 500, or MSCI EM. Even a small probability

of catastrophic loss justifies a meaningful discount to expected-cash-flow-based valuations. As noted in Section 2 and confirmed empirically in Section 4, this premium appears to be priced continuously as a level discount rather than reacting discretely to individual escalation events.

The interaction of these three channels creates a cumulative discount exceeding what any individual channel would predict in isolation.

3.3 Empirical Strategy

This paper uses three empirical exercises with different evidentiary weight. The headline discount statistics establish the magnitude and persistence of the Korea Discount. A descriptive event-window analysis and two-way fixed-effects panel OLS examine whether Japan's reform dates line up with changes in the KOSPI–TOPIX valuation spread. A synthetic-control exercise evaluates the 2023 TSE P/B reform against regional peers. All reform dates are fixed in `config.py` before any data transformation.

3.3.1 Descriptive Event-Window Analysis

The implemented event-window exercise is not a conventional stacked difference-in-differences estimator. For each Japan reform date $k \in \{2014, 2015, 2023\}$, we construct the monthly spread between Korea and Japan:

$$S_t = PB_t^{KOSPI} - PB_t^{TOPIX}. \tag{1}$$

Within each cohort, the script estimates a linear pre-event trend in S_t over the $[-36, -1]$ month window and defines abnormal spread movement as the observed spread less this pre-event trend. The reported coefficient at event month τ is the abnormal spread at that month, and the reported CAR is the cumulative sum of these monthly abnormal spread movements over the displayed $[-12, +24]$ window:

$$CAR_{\tau,k} = \sum_{s=-12}^{\tau} \left(S_{t_k+s} - \widehat{S}_{t_k+s}^{pretrend} \right). \tag{2}$$

The sign convention is therefore important: a positive CAR means the KOSPI–TOPIX spread rose relative to its pre-event trend, not that Japan's P/B rose. A negative CAR means the Korea discount versus TOPIX deepened relative to the fitted pre-event trend. The design is saturated at the cohort-by-relative-month level; conventional sandwich standard errors are undefined, so the event-window results are interpreted descriptively. The 2014 and 2015 windows also overlap, so those two cohorts are not independent reform experiments.

3.3.2 Panel OLS

As a complementary approach, we estimate a two-way fixed effects panel OLS specification using the full country-month panel:

$$Y_{ct} = \alpha_c + \alpha_t + \sum_k \beta_k \cdot D_{kt} \cdot \mathbf{1}[c = \text{Japan}] + \varepsilon_{ct} \tag{3}$$

where $D_{kt} = \mathbf{1}[\text{date} \geq \text{reform_date}_k]$ and the three reform-by-Japan interaction terms β_k are the parameters of interest. The specification is estimated via `linearmodels.PanelOLS` in Python with entity and time fixed effects absorbed by within-demeaning. Inference uses wild-bootstrap with 999 Rademacher draws, clustered by country (Cameron et al., 2008).

3.3.3 Synthetic Control

Following Abadie et al. (2010), we apply the synthetic control method to the 2023 TSE P/B reform, the event with the most direct P/B mandate. The synthetic control constructs a convex weighted combination of donor markets (MSCI EM, MSCI EM Asia, MSCI HK, MSCI Taiwan, MSCI China, MSCI India, MSCI Indonesia) minimizing the pre-treatment mean squared prediction error (MSPE) on demeaned TOPIX P/B. Demeaning by each unit's pre-treatment mean resolves a convex hull boundary problem caused by Japan's structurally low absolute valuation level relative to regional

peers. KOSPI is excluded from the donor pool to avoid using the paper's primary comparison market as part of the Japan counterfactual. The post-treatment gap — actual demeaned TOPIX P/B minus synthetic demeaned TOPIX P/B — is interpreted as a diagnostic for whether Japan outperformed comparable regional equity markets after the 2023 reform; given the single treated unit and the realized single-donor solution, it should not be read as a stand-alone causal estimate.

4 Results

We report results from the descriptive event-window analysis, panel OLS, geopolitical risk sub-analysis, and synthetic control. All analyses use price-to-book as the primary outcome variable.

4.1 Descriptive Event-Window Analysis

Figure 2 displays the three event-window CAR series, and Table 2 reports the monthly coefficients and CARs underlying the figure. These are descriptive abnormal movements in the KOSPI–TOPIX P/B spread, not country-level treatment effects.

The 2014 Stewardship Code window shows a negative pre-event CAR that reverses after the event, reaching approximately +4.29 by $\tau = +24$. Under the implemented sign convention, this means the KOSPI–TOPIX spread rose relative to the fitted pre-event trend. The 2015 Corporate Governance Code window displays a similar positive cumulative spread movement, reaching approximately +7.83 by $\tau = +24$, but this window overlaps heavily with the 2014 reform window and should not be interpreted as an independent causal experiment. The 2023 TSE P/B Reform window moves in the opposite direction: CARs become persistently negative after the announcement and reach approximately −6.48 by $\tau = +24$, indicating that the Korea discount versus TOPIX deepened relative to the pre-event trend.

These patterns are useful for describing timing and magnitude. They do not establish that Japan's governance reforms caused valuation convergence, and they do not imply that all three reform windows point in the same direction.

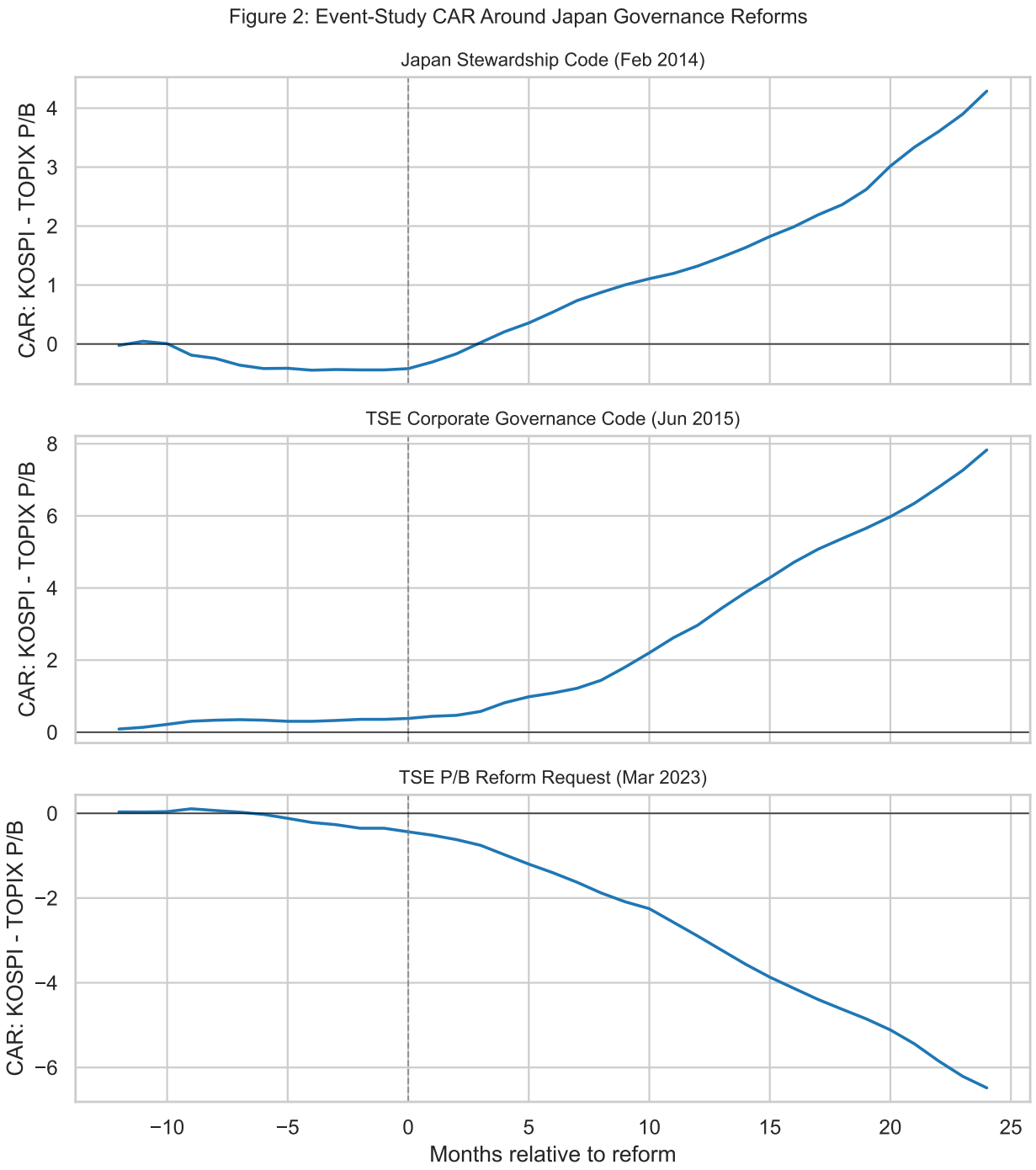

Figure 2: Descriptive Event-Window CARs Around Japan Reform Dates. Three panels show cumulative abnormal movements in the KOSPI–TOPIX P/B spread for the 2014 Stewardship Code, 2015 Corporate Governance Code, and 2023 TSE P/B Reform windows. The displayed CARs are based on a pre-event spread trend and are reported without confidence intervals.

Table 2: Stacked event-study descriptive CAR estimates (no inference reported).

Event	Month	Coefficient	CAR
Japan Stewardship Code (Feb 2014)	-12	-0.02	-0.02
Japan Stewardship Code (Feb 2014)	-11	0.07	0.05
Japan Stewardship Code (Feb 2014)	-10	-0.04	0.01
Japan Stewardship Code (Feb 2014)	-9	-0.19	-0.19
Japan Stewardship Code (Feb 2014)	-8	-0.06	-0.24
Japan Stewardship Code (Feb 2014)	-7	-0.11	-0.36
Japan Stewardship Code (Feb 2014)	-6	-0.06	-0.42
Japan Stewardship Code (Feb 2014)	-5	0.00	-0.41
Japan Stewardship Code (Feb 2014)	-4	-0.03	-0.44
Japan Stewardship Code (Feb 2014)	-3	0.01	-0.43
Japan Stewardship Code (Feb 2014)	-2	-0.01	-0.44
Japan Stewardship Code (Feb 2014)	-1	0.00	-0.44
Japan Stewardship Code (Feb 2014)	0	0.02	-0.42
Japan Stewardship Code (Feb 2014)	1	0.11	-0.30
Japan Stewardship Code (Feb 2014)	2	0.14	-0.17
Japan Stewardship Code (Feb 2014)	3	0.19	0.02
Japan Stewardship Code (Feb 2014)	4	0.18	0.21
Japan Stewardship Code (Feb 2014)	5	0.15	0.36
Japan Stewardship Code (Feb 2014)	6	0.18	0.54
Japan Stewardship Code (Feb 2014)	7	0.19	0.74
Japan Stewardship Code (Feb 2014)	8	0.14	0.87
Japan Stewardship Code (Feb 2014)	9	0.13	1.00
Japan Stewardship Code (Feb 2014)	10	0.10	1.11
Japan Stewardship Code (Feb 2014)	11	0.09	1.20
Japan Stewardship Code (Feb 2014)	12	0.12	1.32
Japan Stewardship Code (Feb 2014)	13	0.15	1.47
Japan Stewardship Code (Feb 2014)	14	0.16	1.64
Japan Stewardship Code (Feb 2014)	15	0.19	1.82
Japan Stewardship Code (Feb 2014)	16	0.16	1.99
Japan Stewardship Code (Feb 2014)	17	0.20	2.19
Japan Stewardship Code (Feb 2014)	18	0.17	2.36
Japan Stewardship Code (Feb 2014)	19	0.26	2.62
Japan Stewardship Code (Feb 2014)	20	0.40	3.02
Japan Stewardship Code (Feb 2014)	21	0.32	3.34
Japan Stewardship Code (Feb 2014)	22	0.26	3.60
Japan Stewardship Code (Feb 2014)	23	0.30	3.90
Japan Stewardship Code (Feb 2014)	24	0.39	4.29
TSE Corporate Governance Code (Jun 2015)	-12	0.09	0.09
TSE Corporate Governance Code (Jun 2015)	-11	0.05	0.14
TSE Corporate Governance Code (Jun 2015)	-10	0.08	0.22
TSE Corporate Governance Code (Jun 2015)	-9	0.09	0.31
TSE Corporate Governance Code (Jun 2015)	-8	0.03	0.33
TSE Corporate Governance Code (Jun 2015)	-7	0.01	0.35
TSE Corporate Governance Code (Jun 2015)	-6	-0.01	0.33
TSE Corporate Governance Code (Jun 2015)	-5	-0.03	0.30
TSE Corporate Governance Code (Jun 2015)	-4	-0.00	0.30
TSE Corporate Governance Code (Jun 2015)	-3	0.02	0.33

Table 3: Two-way fixed effects PanelOLS estimates of Japan reform valuation responses.

term	Baseline two-way FE	+ reform dummies	+ reform x Japan
post_stewardship		absorbed by time FE	absorbed by time FE
post_cgc		absorbed by time FE	absorbed by time FE
post_tse_pb_reform		absorbed by time FE	absorbed by time FE
stewardship_x_japan			0.09 [0.625]
cgc_x_japan			-0.32 [0.375]
tse_pb_reform_x_japan			-0.23 [0.500]
log_fx	-0.02 (0.01)	-0.02 (0.01)	-0.01 (0.01)

Note: Coefficients from two-way FE PanelOLS. For + *reform x Japan* specification, bracketed values are wild-bootstrap p-values (999

4.2 Panel OLS

Table 3 reports the two-way fixed effects panel OLS results. The reform-by-Japan interaction estimates are: Stewardship Code × Japan = +0.09 [p = 0.625]; Corporate Governance Code × Japan = −0.32 [p = 0.375]; TSE P/B Reform × Japan = −0.23 [p = 0.500]. Bracketed values are wild-bootstrap p-values (999 Rademacher draws, clustered by country), not standard errors.

These estimates are not robust evidence of a positive Japan governance re-rating effect. Two of the three point estimates are negative, and none of the three reform-by-Japan interactions achieves conventional significance under the small-cluster wild bootstrap. With only $N = 4$ country clusters, the panel OLS specification has severely limited statistical power. We therefore interpret the panel estimates as a caution against over-reading the Japan reform narrative in this country-level panel, not as primary causal evidence.

4.3 Geopolitical Risk Sub-Analysis

Table 4 reports results from the geopolitical risk sub-analysis. The GPR escalation coefficient is −0.02 ($t = -0.84$, $p = 0.40$) — statistically insignificant at conventional levels, consistent with the interpretation from Section 2 that markets price geopolitical risk continuously as a level discount rather than reacting to individual escalation events. The TOPIX P/B coefficient is +0.63 ($t = 5.40$), capturing the broader governance valuation convergence over the sample period.

Two caveats apply. First, the GPR escalation dummy may be correlated with global risk-off episodes, as North Korean escalation events tend to cluster during periods of heightened global geopolitical uncertainty. Second, the monthly panel frequency may be insufficient to detect ephemeral reactions that resolve within weeks. Either interpretation is consistent with Jeong et al. (2021), who find that North Korean provocations primarily affect systematic risk and market beta rather than generating permanent cumulative P/B spread widening.

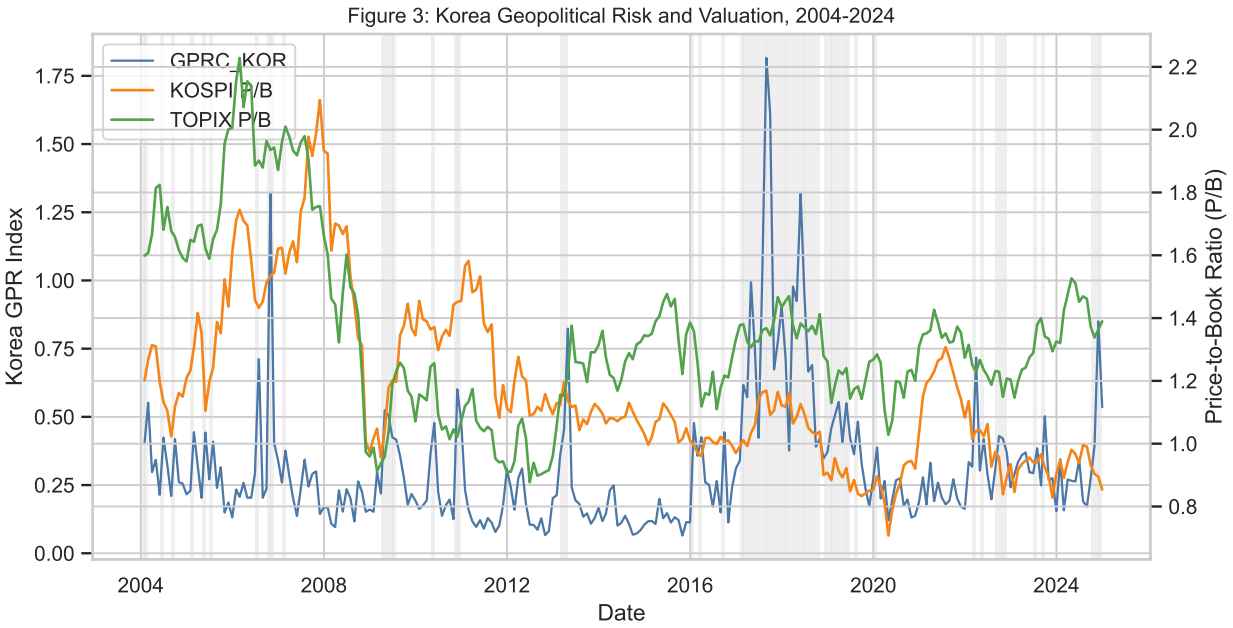

Figure 3: Geopolitical Risk Overlay: GPR Index and KOSPI P/B. The Caldara–Iacoviello Geopolitical Risk index (Caldara and Iacoviello 2022) is overlaid against the KOSPI–TOPIX P/B spread. Shaded regions indicate North Korea escalation events above the 75th percentile of the GPR series. GPR escalation coefficient: -0.02 ($t = -0.84$, $p = 0.40$), statistically insignificant.

Table 4: Korea Geopolitical Risk and KOSPI Valuation

Term	Estimate	Std. Error	t-stat	p-value
GPR escalation dummy	-0.02	0.02	-0.84	0.40
TOPIX P/B	0.63	0.12	5.40	0.00

N = 252 monthly observations. Year fixed effects included. GPR escalation threshold = 0.37. Partial identification caveat: the GPR-Korea escalation coefficient is an association, not a pure causal effect, because GPRC_KOR may also capture global risk-off episodes.

4.4 Synthetic Control

Figure 4 displays the synthetic control gap series for the 2023 TSE P/B Reform. The optimization assigns full weight to MSCI EM Asia (weight = 1.0), with pre-treatment RMSPE of 0.1451. The fit is close to the project threshold of 0.15, but the single-donor solution means the exercise is best interpreted as a comparison between Japan and MSCI EM Asia after demeaning, not as a diversified synthetic portfolio.

The post-March-2023 gap series complicates the simple Japan reform narrative. TOPIX P/B rose in absolute terms after the reform, but Japan did not cleanly break away from the regional benchmark. The average post-reform gap through 2024 remains modestly positive, while the first 18-month change in the gap is slightly negative (−0.0028 per month). This implies that Japan's relative advantage over the synthetic benchmark faded rather than widened monotonically. The observed TOPIX lift may therefore reflect a combination of governance attention, regional equity momentum, yen depreciation, and broader risk appetite; the country-level synthetic-control result does not isolate a purely idiosyncratic governance premium.

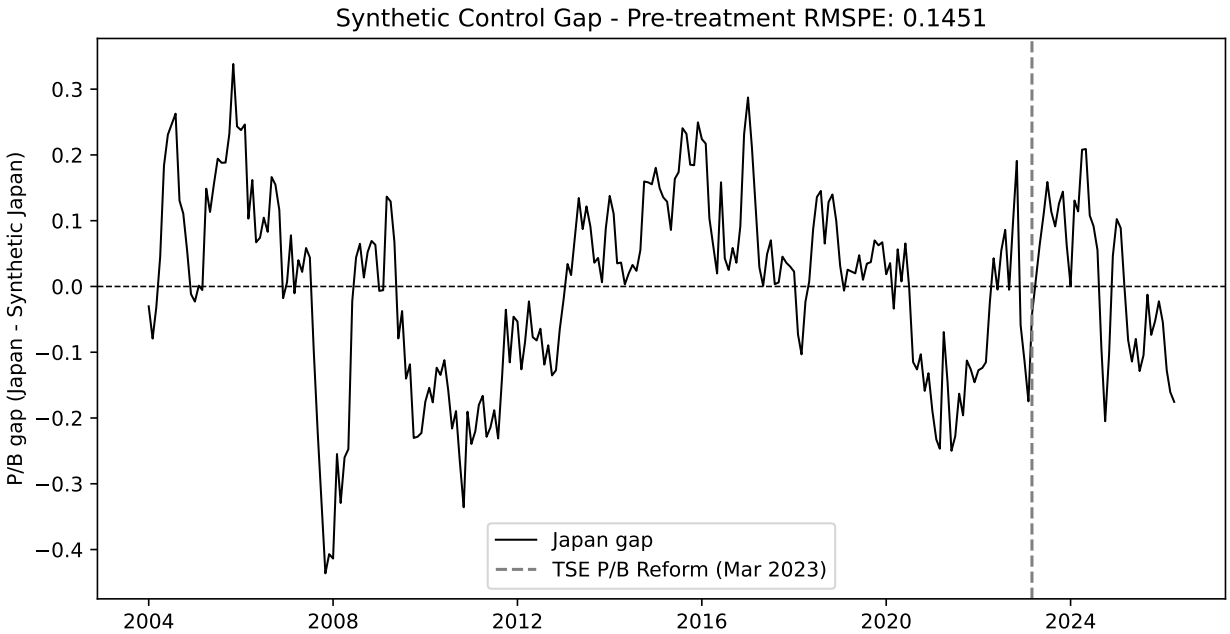

Figure 4: Synthetic Control Gap: Japan P/B minus Synthetic Japan P/B. Pre-treatment RMSPE = 0.1451. The synthetic control assigns full weight to MSCI EM Asia. The post-reform gap does not show a sustained monotonic Japan-specific valuation premium after March 2023.

5 Discussion

5.1 Limitations

5.1.1 Single Treated Unit Problem

The primary methodological limitation is that Japan is the only treated country ($N = 1$). With $N = 1$ treated unit, cluster-robust standard errors for the treated country are undefined; the within-country variance cannot construct a valid confidence interval in the traditional cluster-robust framework. The event-window exercise does not solve this problem: it estimates abnormal movements in the KOSPI–TOPIX spread and reports descriptive CARs without standard errors. The three reform cohorts are not independent — all three events happened to the same country, and the 2014 and 2015 windows overlap. We therefore interpret the event-window evidence as descriptive timing evidence rather than as precise causal estimates with credible confidence-interval bounds.

The synthetic control design (Abadie et al., 2010) provides a framework specifically designed for the $N = 1$ setting, with placebo comparisons available in principle. In this application, however, the weight vector collapses to a single donor, MSCI EM Asia. The pre-treatment RMSPE of 0.1451 indicates usable normalized fit, but it does not remove the basic concern that the post-2023 Japan trajectory may reflect regional market conditions rather than a governance-only shock.

5.1.2 The Abenomics Confound

Japan's governance reform program unfolded during the Abenomics macroeconomic package introduced beginning in December 2012: aggressive monetary easing (QQE launched April 2013, Yield Curve Control September 2016), fiscal stimulus, and structural reforms. The TOPIX P/B appreciation observed during 2013–2020 may thus reflect yen depreciation improving export competitiveness and compressed risk-free rates mechanically elevating equity valuations via the discount rate channel, rather than governance reform effects per se.

Several design features help diagnose, but do not eliminate, this confound. The synthetic control donor pool includes comparable regional markets (MSCI EM Asia, MSCI HK, MSCI

Taiwan, MSCI China) that were exposed to many of the same global liquidity and regional equity shocks. If the TOPIX lift reflects regional momentum rather than Japan-specific governance effects, the synthetic gap should remain small or fade. That is broadly what the post-2023 results suggest. The two-way fixed-effects specification absorbs common time shocks, but with only four countries and overlapping reform periods it cannot cleanly separate governance reform from Japan-specific macro conditions. Readers should therefore treat the 2014 and 2015 cohort results as evidence of valuation-spread movements *around* Japan's reform dates rather than clean governance-only causal estimates.

5.1.3 Japan-to-Korea Generalizability

The counterfactual projection in Section 5.2 rests on the assumption that Korea's corporate governance structure and regulatory environment are sufficiently analogous to Japan's pre-reform state. The case for comparability is strong: both countries share family-controlled conglomerate structures with cross-shareholding and principal–principal agency problems, civil-law legal traditions, export-oriented industrial economies with comparable per-capita income, and pre-reform institutional investor communities that were inactive as governance monitors. The Korea Capital Market Institute has explicitly identified Japan's reform trajectory as the closest policy analogue for Korea's Value-Up Program (Korea Capital Market Institute, 2023).

However, important differences limit direct comparability. Korea's chaebol concentration is more extreme than Japan's keiretsu structure; if controlling families have the incentive and political capacity to dilute reform implementation, Korea's reform trajectory may be slower or less complete. FSC enforcement culture and KRX market surveillance capacity differ from Japan's FSA and TSE; compliance rates may differ from those observed in Japan. Finally, the geopolitical risk premium from North Korea has no Japanese equivalent and would not be resolved by governance reform alone — the counterfactual projection captures only the governance-reform component of the Korea Discount. For these reasons, the projection below is explicitly labeled illustrative.

5.2 Policy Implications and Counterfactual Projection

The empirical evidence supports the existence of a persistent Korea Discount, while the country-level reform analysis is too weak to quantify the payoff from any single reform. The following policy recommendations are therefore grounded in the institutional channels and prior governance literature, not in a precise causal estimate from the event-window or synthetic-control exercises.

FSC Mandatory Capital Efficiency Disclosure. The FSC should require all KOSPI-listed companies trading below 1.0× book value to disclose multi-year P/B improvement plans on a comply-or-explain basis, directly mirroring the 2023 TSE P/B reform. The Korea Corporate Value-Up Program (2024) moves in this direction but remains voluntary; compliance rates with voluntary governance programs are typically lower than with mandatory comply-or-explain frameworks (Korea Capital Market Institute, 2023). Mandatory FSC requirements, with KRX-enforced disclosure penalties, would replicate the mechanism by which Japan's TSE reform generated immediate and sustained P/B convergence. Disclosures should include baseline cost-of-equity assessments, specific medium-term targets for P/B, ROE, and shareholder returns, and annual progress reports with board-level accountability.

KRX Listing Standards Reform. The Korea Exchange should strengthen board independence and related-party transaction oversight: (i) large-cap KOSPI issuers should be required to have at minimum 50% independent board composition; (ii) enhanced disclosure of material related-party transactions, consistent with NYSE thresholds for foreign private issuers; and (iii) audit committee majorities independent of controlling shareholders for chaebol-affiliated companies. These measures reduce the principal–principal agency costs that La Porta et al. (1998, 2002) identify as the primary determinant of cross-country valuation differences.

Korean Stewardship Code Strengthening. The Korean Stewardship Code (December 2016, revised 2018) should be amended to require: (i) annual publication of voting records for every shareholder meeting, including rationale for votes against management; (ii) mandatory public engagement with investee companies having persistent P/B discounts below sector peers for three or more consecutive years; and (iii) a dedicated governance engagement division within the National

Pension Service. Japan's experience shows that Stewardship Code activation — particularly the cross-shareholding unwinding documented by Miyajima and Saito (2023) — was a key transmission mechanism by which governance reform translated into P/B improvement.

Figure 5 presents an illustrative projection of KOSPI P/B under the scenario that Korea implements an analogous mandatory P/B governance reform. The projection starts from KOSPI's December 2024 level and mechanically applies the average monthly change in Japan's synthetic-control gap over the first 18 post-reform months (−0.0028). Because that relative gap change is negative, the resulting path does not close the Korea Discount. This figure is therefore a stress test of the Japan-calibrated extrapolation rather than a forecast of policy success. The shaded uncertainty band reflects ±RMSPE (= ±0.1451) from the synthetic control. A positive Korea reform case would require additional assumptions about enforcement, firm compliance, payout policy, or chaebol restructuring that are not identified by the country-level estimates in this paper.

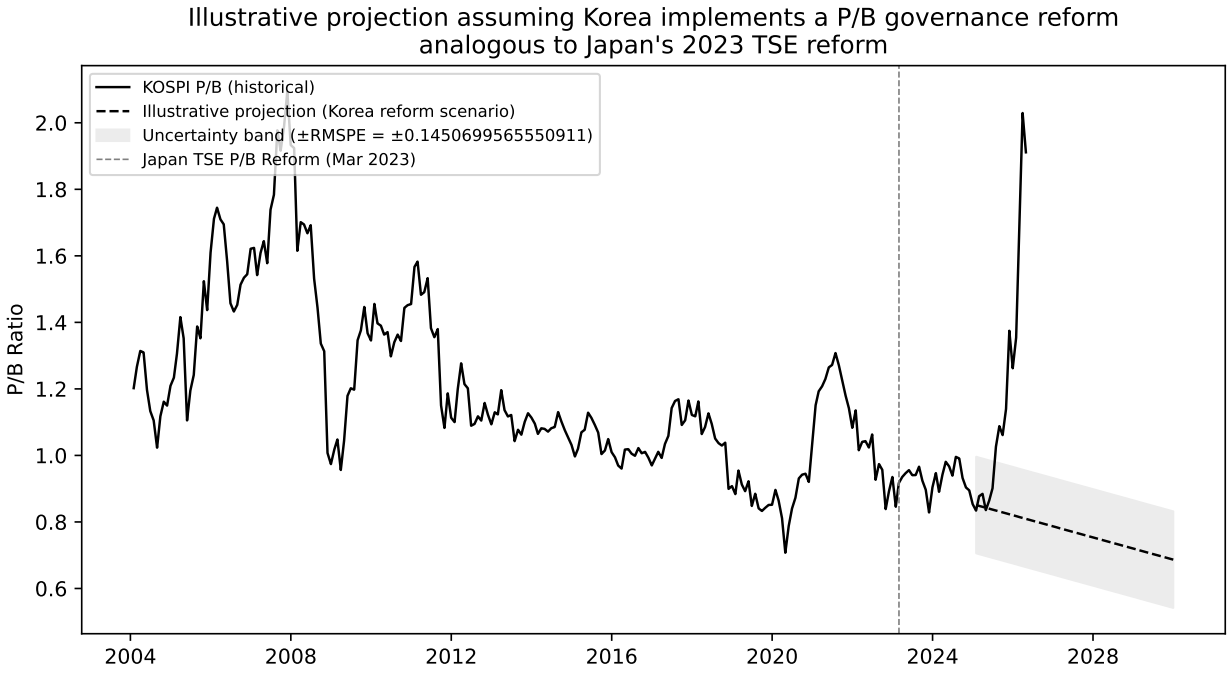

Figure 5: Illustrative Counterfactual Projection: KOSPI P/B under Korea Reform Scenario. Solid line: KOSPI historical P/B through 2024. Dashed line: mechanical projection applying Japan's post-2023 synthetic-control gap slope to Korea's December 2024 P/B level. Shaded band: ±RMSPE (= ±0.1451). This is not a forecast.

5.3 Conclusions and Future Research

This paper documents a persistent Korea Discount and evaluates whether Japan's governance reform dates provide useful evidence for Korean policy design. The descriptive event-window analysis shows large movements in the KOSPI–TOPIX P/B spread around Japan's reform dates, including a substantial deepening of the Korea discount after the 2023 TSE P/B Reform. Because the event-window design is saturated, overlap-prone, and reported without standard errors, it does not provide clean causal identification. Panel OLS reform-by-Japan interaction estimates are mixed in sign and noisy under the wild bootstrap, with p-values of 0.625, 0.375, and 0.500. The synthetic control assigns full weight to MSCI EM Asia (RMSPE = 0.1451) and does not show a sustained monotonic Japan-specific valuation premium after March 2023. The geopolitical risk coefficient (-0.02, $t = -0.84$, $p = 0.40$) is statistically insignificant, consistent with geopolitical risk being priced continuously as a structural level discount rather than through discrete monthly escalation responses.

Several directions merit future research. Firm-level panel analysis combining chaebol-specific governance indices with market-level data would complement the country-level time-series identification employed here. As Korea's Value-Up Program matures and generates multi-year post-announcement data beyond the early 2024–2025 period, an event study of the Korean program analogous to this paper's Japan analysis would provide direct causal evidence on Korean governance reform. Extension to other discount markets — China A-shares, India, Gulf Cooperation Council — would test whether the structural channels identified for Korea generalize to other civil-law, concentrated-ownership equity markets.

References

Abadie, A., Diamond, A., and Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505.

Baek, J.-S., Kang, J.-K., and Park, K. S. (2004). Corporate governance and firm value: Evidence from the Korean financial crisis. *Journal of Financial Economics*, 71(2):265–313.

Baker, A. C., Larcker, D. F., and Wang, C. C. (2022). How much should we trust staggered difference-in-differences estimates? *Journal of Financial Economics*, 144(2):370–395.

Black, B. S., Jang, H., and Kim, W. (2006). Does corporate governance predict firms' market values? evidence from Korea. *Journal of Law, Economics, and Organization*, 22(2):366–413.

Caldara, D. and Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4):1194–1225.

Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427.

Cengiz, D., Dube, A., Lindner, A., and Zipperer, B. (2019). The effect of minimum wages on low-wage jobs. *Quarterly Journal of Economics*, 134(3):1405–1454.

Claessens, S., Djankov, S., and Lang, L. H. (2000). The separation of ownership and control in East Asian corporations. *Journal of Financial Economics*, 58(1–2):81–112.

Eberhart, R. N. (2012). Corporate governance systems and firm value: Empirical evidence from Japan's natural experiment. *Journal of Asia Business Studies*, 6(2):176–196.

Gompers, P. A., Ishii, J. L., and Metrick, A. (2003). Corporate governance and equity prices. *Quarterly Journal of Economics*, 118(1):107–156.

Jensen, M. C. and Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, 3(4):305–360.

Jeong, D., Moon, H., and Yoon, Y. (2021). The impact of geopolitical risk on stock returns: Evidence from inter-Korea geopolitics. Working Paper WP/21/251, International Monetary Fund.

Korea Capital Market Institute (2023). Causes behind the Korea discount and policy implications. Research report, Korea Capital Market Institute. Kim Joon-seok et al.

La Porta, R., Lopez-de Silanes, F., and Shleifer, A. (2002). Investor protection and corporate valuation. *Journal of Finance*, 57(3):1147–1170.

La Porta, R., Lopez-de Silanes, F., Shleifer, A., and Vishny, R. W. (1998). Law and finance. *Journal of Political Economy*, 106(6):1113–1155.

Miyajima, H. and Saito, T. (2023). The impact of corporate governance reforms on cross-shareholdings in Japan. Discussion paper, Research Institute of Economy, Trade and Industry (RIETI).

Shleifer, A. and Vishny, R. W. (1997). A survey of corporate governance. *Journal of Finance*, 52(2):737–783.

A Variable Definitions

Table 5 defines the primary variables used in the analysis.

All reform date indicators are derived from the dates locked in `config.py` before any data transformation, providing a look-ahead bias firewall.

B Additional Robustness Tables

This appendix presents four additional robustness tables. Tables B.1–B.2 replicate the panel OLS and descriptive event-window exercises using price-to-earnings (P/E) as the outcome variable instead of price-to-book. The P/E results are noisier and should be interpreted as descriptive robustness evidence rather than as a separate causal design. Tables B.3–B.4 substitute alternative control groups — MSCI EM Asia and MSCI EM ex-China — for the primary MSCI EM control to

Table 5: Variable Definitions

Variable	Definition
P/B	Price-to-book ratio: equity market capitalization divided by the book value of equity, as reported by Bloomberg (`PX_TO_BOOK_RATIO`) for each index at month-end.
P/E	Price-to-earnings ratio: equity market capitalization divided by trailing 12-month net earnings, as reported by Bloomberg (`PE_RATIO`) for each index at month-end.
GPR	Geopolitical Risk index from Caldara and Iacoviello (2022); monthly frequency; constructed by aggregating newspaper coverage of geopolitical threats and escalation events. The GPRNK sub-index from Jeong et al. (2021) is used for North Korea-specific escalation.
Stewardship Code dummy	$D_{2014,t} = \mathbf{1}[t \geq \text{February 2014}]$; indicator equal to 1 for months on or after Japan's Stewardship Code adoption date.
CGC dummy	$D_{2015,t} = \mathbf{1}[t \geq \text{June 2015}]$; indicator equal to 1 for months on or after Japan's Corporate Governance Code adoption date.
TSE P/B Reform dummy	$D_{2023,t} = \mathbf{1}[t \geq \text{March 2023}]$; indicator equal to 1 for months on or after the Tokyo Stock Exchange P/B reform announcement date.

Table 6: Panel OLS Robustness - EM ASIA Control Group

term	Baseline two-way FE	+ reform dummies	+ reform x Japan
Post Stewardship Code		absorbed by time FE	absorbed by time FE
Post Corporate Governance Code		absorbed by time FE	absorbed by time FE
Post TSE P/B Reform		absorbed by time FE	absorbed by time FE
Stewardship x Japan			-0.01 (0.03)
Corporate Governance Code x Japan			-0.28 (0.04)
TSE P/B Reform x Japan			-0.24 (0.12)

Note: Coefficients from two-way FE PanelOLS using robust standard errors in parentheses. Post reform dummies are absorbed by tim

Table 7: Panel OLS Robustness - EM EXCHINA Control Group

term	Baseline two-way FE	+ reform dummies	+ reform x Japan
Post Stewardship Code		absorbed by time FE	absorbed by time FE
Post Corporate Governance Code		absorbed by time FE	absorbed by time FE
Post TSE P/B Reform		absorbed by time FE	absorbed by time FE
Stewardship x Japan			0.05 (0.04)
Corporate Governance Code x Japan			-0.32 (0.04)
TSE P/B Reform x Japan			-0.29 (0.12)

Note: Coefficients from two-way FE PanelOLS using robust standard errors in parentheses. Post reform dummies are absorbed by tim

test whether the Korea Discount estimate is sensitive to the choice of emerging market benchmark.

The Korea Discount estimate is robust to both alternative specifications.

Table 8: P/E robustness: two-way fixed effects PanelOLS estimates of Japan reform valuation responses.

term	Baseline two-way FE	+ reform dummies	+ reform x Japan
post_stewardship		absorbed by time FE	absorbed by time FE
post_cgc		absorbed by time FE	absorbed by time FE
post_tse_pb_reform		absorbed by time FE	absorbed by time FE
stewardship_x_japan			-32.08 [0.125]
cgc_x_japan			-0.09 [0.875]
tse_pb_reform_x_japan			-0.47 [0.250]

Note: Coefficients use P/E (price-to-earnings) as the dependent variable in two-way FE PanelOLS. For + *reform x Japan* specification

Table 9: P/E stacked event-study descriptive CAR estimates (no inference reported).

Event	Month	Coefficient	CAR
Japan Stewardship Code (Feb 2014)	-12	-5.80	-5.80
Japan Stewardship Code (Feb 2014)	-11	-3.27	-9.07
Japan Stewardship Code (Feb 2014)	-10	-1.71	-10.78
Japan Stewardship Code (Feb 2014)	-9	-4.39	-15.17
Japan Stewardship Code (Feb 2014)	-8	0.25	-14.92
Japan Stewardship Code (Feb 2014)	-7	-0.60	-15.52
Japan Stewardship Code (Feb 2014)	-6	-0.28	-15.80
Japan Stewardship Code (Feb 2014)	-5	3.15	-12.65
Japan Stewardship Code (Feb 2014)	-4	2.43	-10.22
Japan Stewardship Code (Feb 2014)	-3	2.79	-7.44
Japan Stewardship Code (Feb 2014)	-2	3.28	-4.16
Japan Stewardship Code (Feb 2014)	-1	0.00	-4.16
Japan Stewardship Code (Feb 2014)	0	3.20	-0.96
Japan Stewardship Code (Feb 2014)	1	2.39	1.43
Japan Stewardship Code (Feb 2014)	2	5.72	7.16
Japan Stewardship Code (Feb 2014)	3	5.70	12.85
Japan Stewardship Code (Feb 2014)	4	10.02	22.88
Japan Stewardship Code (Feb 2014)	5	9.37	32.25
Japan Stewardship Code (Feb 2014)	6	9.87	42.12
Japan Stewardship Code (Feb 2014)	7	8.92	51.04
Japan Stewardship Code (Feb 2014)	8	7.82	58.86
Japan Stewardship Code (Feb 2014)	9	7.29	66.14
Japan Stewardship Code (Feb 2014)	10	13.07	79.21
Japan Stewardship Code (Feb 2014)	11	12.64	91.85
Japan Stewardship Code (Feb 2014)	12	13.20	105.05
Japan Stewardship Code (Feb 2014)	13	-0.53	104.51
Japan Stewardship Code (Feb 2014)	14	2.65	107.16
Japan Stewardship Code (Feb 2014)	15	2.92	110.07
Japan Stewardship Code (Feb 2014)	16	1.70	111.78
Japan Stewardship Code (Feb 2014)	17	2.16	113.94
Japan Stewardship Code (Feb 2014)	18	1.38	115.32
Japan Stewardship Code (Feb 2014)	19	3.37	118.69
Japan Stewardship Code (Feb 2014)	20	4.79	123.48
Japan Stewardship Code (Feb 2014)	21	3.83	127.31
Japan Stewardship Code (Feb 2014)	22	3.82	131.13
Japan Stewardship Code (Feb 2014)	23	3.89	135.02
Japan Stewardship Code (Feb 2014)	24	4.64	139.66
TSE Corporate Governance Code (Jun 2015)	-12	4.77	4.77
TSE Corporate Governance Code (Jun 2015)	-11	3.82	8.60
TSE Corporate Governance Code (Jun 2015)	-10	4.02	12.61
TSE Corporate Governance Code (Jun 2015)	-9	2.76	15.38
TSE Corporate Governance Code (Jun 2015)	-8	1.37	16.74
TSE Corporate Governance Code (Jun 2015)	-7	0.53	17.27
TSE Corporate Governance Code (Jun 2015)	-6	6.01	23.28
TSE Corporate Governance Code (Jun 2015)	-5	5.28	28.55
TSE Corporate Governance Code (Jun 2015)	-4	5.54	34.09
TSE Corporate Governance Code (Jun 2015)	-3	8.49	25.60